Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 4 — Solutions

October 2, 2026

Exercise 1. Let $A = \{x \mid x > 0\}$, $B = \{x \mid x \neq 1\}$, and $C = \{x \mid x < 1\}$.
Is the following equality correct?

$$(A \cap B) \cup C = A \cup (B \cap C)$$

Let's analyze each set individually and then compare them:
Step 1: Find $(A \cap B) \cup C$:

$$A \cap B = \{x \mid x > 0\} \cap \{x \mid x \neq 1\} = \{x \mid x > 0 \text{ and } x \neq 1\}$$

thus,

$$(A \cap B) \cup C = \{x \mid x > 0 \text{ and } x \neq 1\} \cup \{x \mid x < 1\} = \{x \mid x \neq 1\}$$

Step 2: Find $A \cup (B \cap C)$

$$B \cap C = \{x \mid x \neq 1\} \cap \{x \mid x < 1\} = C$$

thus,

$$A \cup (B \cap C) = \{x \mid x > 0\} \cup \{x \mid x < 1\} = \mathbf{R}$$

After simplification, we can see that $(A \cap B) \cup C \neq A \cup (B \cap C)$, hence the equality is not correct.

Exercise 2. Let $E = A \cap (B - C)$ and $F = (A \cap B) - (A \cap C)$ Which of the following propositions is always true?

- ✓ $E \subseteq F$
- ✓ $F \subseteq E$
- ✓ $E = F$
- ◯ None of the other answers are always true.

Hint: $A - B = A \cap \overline{B}$
Let's break down the sets E and F:

$$\begin{aligned} E &= A \cap (B - C) \\ &= A \cap (B \cap \overline{C}) \text{ (from the hint)} \\ &= (A \cap B) \cap \overline{C} \text{ (associative law)} \end{aligned}$$

$$\begin{aligned} F &= (A \cap B) - (A \cap C) \\ &= (A \cap B) \cap \overline{(A \cap C)} \text{ (from the hint)} \\ &= (A \cap B) \cap (\overline{A} \cup \overline{C}) \text{ (De Morgan's Laws)} \\ &= (A \cap B \cap \overline{A}) \cup (A \cap B \cap \overline{C}) \text{ (Distributive law)} \end{aligned}$$

Because $(A \cap B \cap \overline{A}) = \emptyset$, we have

$$F = (A \cap B \cap \overline{C})$$

Hence $E = F$, which also implies that $E \subseteq F$ and $F \subseteq E$.

Exercise 3. Let $f : \{x \mid x \in \mathbf{R}, -2 \leq x \leq 5\} \to \mathbf{R}$,

$$x \mapsto \begin{cases} 3 + \frac{3}{2}x & \text{for} \quad -2 \leq x \leq 0 \\ \lfloor x \rfloor & \text{for} \quad 0 \leq x < 2 \\ x^2 & \text{for} \quad 2 \leq x \leq 5. \end{cases}$$

◯ f is injective but not surjective.

◯ f is surjective but not injective.

◯ f is bijective.

◯ f is not a function.

For $-2 \leq x \leq 0$ we have that x is mapped to $3 + \frac{3}{2}x$, which equals 3 for $x = 0$. But for $0 \leq x < 2$ we have that x is mapped to $\lfloor x \rfloor$, which equals 0 for $x = 0$. Thus $x = 0$ is mapped by f to both 3 and to 0, which implies that the definition of f violates the definition of a function, namely that each value of the domain has a single function value.

It follows that the last circle must be ticked.

Exercise 4. Let $f : \{x \mid x \in \mathbf{R}, 0 < x < 1\} \to \mathbf{R}$,

$$x \mapsto \begin{cases} 2 - \dfrac{1}{x} & \text{if} \quad 0 < x < 1/2 \\ \dfrac{1}{1-x} - 2 & \text{if} \quad 1/2 \leq x < 1. \end{cases}$$

◯ f is not injective and not surjective.

◯ f is injective but not surjective.

◯ f is surjective but not injective.

◯ f is bijective.

Proving injectivity: $\forall x_1, x_2 \; f(x_1) = f(x_2) \to x_1 = x_2$.
Assume $f(x_1) = f(x_2)$ and consider every possible combination of x_1, x_2.

- $0 < x_1, x_2 < 1/2$.

  $2 - \frac{1}{x_1} = 2 - \frac{1}{x_2}$ obviously implies $x_1 = x_2$.

- $0 < x_1 < 1/2$ and $1/2 \leq x_2 < 1$ (or $1/2 \leq x_1 < 1$ and $0 < x_2 < 1/2$).

  In this case,

$$\begin{aligned} 2 - \frac{1}{x_1} &= \frac{1}{1-x_2} - 2 \\ \frac{1}{1-x_2} + \frac{1}{x_1} &= 4. \end{aligned}$$

However, since $0 < x_1 < 1/2$ and $1/2 \leq x_2 < 1$, we have

$$\begin{cases} \dfrac{1}{x_1} & > & 2 \\ \dfrac{1}{1-x_2} & \geq & 2. \end{cases}$$

Hence, $\frac{1}{x_1} + \frac{1}{1-x_2} > 4$, so $f(x_1) \neq f(x_2)$ (which still makes the statement true).

- $1/2 \leq x_1, x_2 < 1$.

  $\frac{1}{1-x_1} - 2 = \frac{1}{1-x_2} - 2$ obviously implies $x_1 = x_2$.

Therefore, $f(x)$ is *injective*.

Proving surjectivity: $\forall y \, \exists x \; f(x) = y$.
Let $y \in \mathbf{R}$ and consider the two following cases.

- $y < 0$.

  In this case, for $0 < x < 1/2$, we have

$$\begin{aligned} 2 - \frac{1}{x} &= y \\ x(2-y) &= 1 \\ x &= \frac{1}{2-y} \end{aligned}$$

  which respects the constraints on x and, hence, validates the statement.

- $y \geq 0$.

  In this case, for $1/2 \leq x < 1$, we have

$$\begin{aligned} \frac{1}{1-x} - 2 &= y \\ 1 - 2(1-x) &= y(1-x) \\ x(y+2) &= y+1 \\ x &= \frac{y+1}{y+2} \end{aligned}$$

  which respects the constraints on x and, hence, validates the statement.

Therefore, $f(x)$ is *surjective*.
$f(x)$ is surjective and injective. Therefore it is bijective !!!

Exercise 5. Determine which of these functions are bijections from $\mathbf{R}$ to $\mathbf{R}$

- ◯ $f(x) = 2x + 1$
- ◯ $f(x) = x^2 + 1$
- ◯ $f(x) = x^3$
- ◯ $f(x) = (x^2 + 1)/(x^2 + 2)$

1. To show that the function is one-to-one, note that if $2x + 1 = 2x' + 1$, then $x = x'$. To show that the function is onto, note that $2((y - 1)/2) + 1 = y$, so every number is in the range.

2. This function is not a bijection, since its range is the set of real numbers greater than or equal to 1 (which is sometimes written $[1, \infty)$), not all of $\mathbf{R}$. (It is not injective either.)

3. This function is a bijection, since it has an inverse function, namely the function $f(y) = y^{1/3}$ (obtained by solving $y = x^3$ for x).

4. This function is not a bijection. It is easy to see that it is not injective, since x and -x have the same image, for all real numbers x. A little work shows that the range is only $\{y | 0.5 \leq y < 1\} = [0.5, 1)$.

Exercise 6. For any $\delta \in \mathbf{R}$ let f_δ and g_δ be the following two functions from $\mathbf{R}$ to $\mathbf{R}$

$$f_\delta(x) = \begin{cases} x + \delta & \text{if } x \in \mathbf{Z} \\ -x + \delta & \text{if } x \notin \mathbf{Z}, \end{cases} \qquad g_\delta(x) = \begin{cases} x + \delta & \text{if } x \in \mathbf{Z} \\ -x - \delta & \text{if } x \notin \mathbf{Z}. \end{cases}$$

Consider the two statements

$$\forall \delta \in \mathbf{R} \;\; f_\delta \text{ is a bijection} \qquad \text{and} \qquad \forall \delta \in \mathbf{R} \;\; g_\delta \text{ is a bijection.}$$

◯ Only the second statement is true.

◯ Only the first statement is true.

◯ They are both true.

◯ They are both false.

Proving injectivity: $\forall x_1, x_2 \; f(x_1) = f(x_2) \to x_1 = x_2$.

- Given any distinct $x_1, x_2 \in \mathbf{R}$, if $x_1 \in \mathbf{Z}$ and $x_2 \in \mathbf{Z}$, then $f_\delta(x_1) = x_1 + \delta \neq x_2 + \delta = f_\delta(x_2)$, otherwise if $x_1 \in \mathbf{Z}$ and $x_2 \notin \mathbf{Z}$, then also $-x_2 \notin \mathbf{Z}$ so that $x_1 \neq -x_2$ and $f_\delta(x_1) = x_1 + \delta \neq -x_2 + \delta = f_\delta(x_2)$, and otherwise if $x_1 \notin \mathbf{Z}$ and $x_2 \notin \mathbf{Z}$, then $f_\delta(x_1) = -x_1 + \delta \neq -x_2 + \delta = f_\delta(x_2)$.

  Because the remaining case $x_1 \notin \mathbf{Z}$ and $x_2 \in \mathbf{Z}$ is equivalent to the case $x_1 \in \mathbf{Z}$ and $x_2 \notin \mathbf{Z}$ that was treated already, we find that for any $x_1 \neq x_2$ we have that $f_\delta(x_1) \neq f_\delta(x_2)$.

  This argument works for any δ.

- To find a counterexample, take one integer input, say $x_1 = 0$, and require $g_\delta(0) = g_\delta(x_2)$ for some $x_2 \notin \mathbf{Z}$. This gives $\delta = -x_2 - \delta$, hence $x_2 = -2\delta$. Choosing, for example, $\delta = \frac{1}{3}$ gives $x_2 = -\frac{2}{3} \notin \mathbf{Z}$. Therefore

  $$g_{(1/3)}(0) = \frac{1}{3} = \frac{2}{3} - \frac{1}{3} = g_{(1/3)}\left(-\frac{2}{3}\right),$$

  so $g_{(1/3)}$ is not injective and therefore not bijective.

Proving surjectivity: $\forall y \, \exists x \; f(x) = y$.

- Given an arbitrary $y \in \mathbf{R}$, if $y - \delta \in \mathbf{Z}$ then $f_\delta(y - \delta) = (y - \delta) + \delta = y$,
  and otherwise if $y - \delta \notin \mathbf{Z}$ then $-(y - \delta) \notin \mathbf{Z}$ and $f_\delta(-(y - \delta)) = -(-(y - \delta)) + \delta = (y - \delta) + \delta = y$.
  This argument works for any δ.

Based on the above, it follows that for any δ the function f_δ is bijective while the function g_δ is not bijective. The negation of the second statement (namely "$\exists \delta \in \mathbf{R}$ g_δ is not a bijection") is correct, implying that the second statement must be ticked. (Note that there exists at least one value for δ such that g_δ is bijective, namely $\delta = 0$ because $f_0 = g_0$.)

Other correction :

Is f_δ injective ?

We need to check that every element in the domain of f_δ is assigned to a different element in the codomain of f_δ.

More formally, we need to check that $\forall x_1, x_2 \; f_\delta(x_1) = f_\delta(x_2) \to x_1 = x_2$. To do so we have to consider 4 cases:

1. $x_1 \in \mathbf{Z}$ and $x_2 \in \mathbf{Z}$
2. $x_1 \notin \mathbf{Z}$ and $x_2 \notin \mathbf{Z}$
3. $x_1 \notin \mathbf{Z}$ and $x_2 \in \mathbf{Z}$
4. $x_1 \in \mathbf{Z}$ and $x_2 \notin \mathbf{Z}$

Let's check all cases:

1. $f_\delta(x_1) = f_\delta(x_2) \rightarrow x_1 + \delta = x_2 + \delta \rightarrow x_1 = x_2$
2. $f_\delta(x_1) = f_\delta(x_2) \rightarrow -x_1 + \delta = -x_2 + \delta \rightarrow x_1 = x_2$
3. $f_\delta(x_1) = f_\delta(x_2) \rightarrow -x_1 + \delta = x_2 + \delta \rightarrow -x_1 = x_2$ This last equality is not possible:
   - We are in the case where x_2 is an integer. Therefore $-x_2$ is an integer as well.
   - However, $-x_2 = x_1$ and we are in the case where x_1 is not an integer: there is a contradiction.
   - By contradiction $f_\delta(x_1) = f_\delta(x_2)$ is false in this case.
   - Therefore the implication $f_\delta(x_1) = f_\delta(x_2) \rightarrow x_1 = x_2$ is true in this case.
4. Same as case 3, we just need to swap x_1 and x_2.

f_δ is injective !!!

Is f_δ surjective ?

We need to check that every element y in the codoamin of f_δ has a preimage x (or inverse image) in the domain of f_δ.

More formally, we need to check that $\forall y \exists x \ f_\delta(x) = y$

Let us pick an arbitrary $y \in \mathbf{R}$ and an arbitrary $\delta \in \mathbf{R}$. If we can show that there is an $x \in \mathbf{R}$ such that $f_\delta(x) = y$ for any values of y and δ, it means that the f_δ is surjective.

Let us start by stating $f_\delta(x) = y$ and let us try to find a value of x for every possible value of y and δ:

$$f_\delta(x) = y \rightarrow \begin{cases} x + \delta = y & \text{if } x \in \mathbf{Z} \\ -x + \delta = y & \text{if } x \notin \mathbf{Z} \end{cases} \rightarrow \begin{cases} x = y - \delta & \text{if } x \in \mathbf{Z} \\ x = \delta - y & \text{if } x \notin \mathbf{Z} \end{cases} \rightarrow \begin{cases} x = y - \delta & \text{if } y - \delta \in \mathbf{Z} \\ x = \delta - y & \text{if } \delta - y \notin \mathbf{Z} \end{cases}$$

We can conclude that for any y and any δ, we can find x such that $f_\delta(x) = y$:
If $y - \delta \in \mathbf{Z}$, we need to set $x = y - \delta$ and otherwise, if $y - \delta \notin \mathbf{Z}$[1], we need to set $x = \delta - y$

f_δ is surjective and injective. Therefore it is bijective !!!

Is g_δ injective ?

We need to check that every element in the domain of g_δ is assigned to a different element in the codomain of g_δ.

More formally, we need to check that $\forall x_1, x_2 \ g_\delta(x_1) = g_\delta(x_2) \rightarrow x_1 = x_2$. To do so we have to consider 4 cases:

1. $x_1 \in \mathbf{Z}$ and $x_2 \in \mathbf{Z}$
2. $x_1 \notin \mathbf{Z}$ and $x_2 \notin \mathbf{Z}$
3. $x_1 \notin \mathbf{Z}$ and $x_2 \in \mathbf{Z}$
4. $x_1 \in \mathbf{Z}$ and $x_2 \notin \mathbf{Z}$

1. $g_\delta(x_1) = g_\delta(x_2) \rightarrow x_1 + \delta = x_2 + \delta \rightarrow x_1 = x_2$

[1] if $\delta - y \notin \mathbf{Z}$, then $y - \delta \notin \mathbf{Z}$, so we can swap y and δ

2. $g_\delta(x_1) = g_\delta(x_2) \to -x_1 - \delta = -x_2 - \delta \to x_1 = x_2$
3. $g_\delta(x_1) = g_\delta(x_2) \to -x_1 - \delta = x_2 + \delta \to -x_1 = x_2 - 2\delta$. It is not injective[2].
4. Same as case 3, we just need to swap x_1 and x_2.

g_δ is not injective. Therefore it is not bijective !!!

Exercise 7. Let $\mathcal{P}(X)$ denote the power set of a set X and let $\emptyset$ denote the empty set. Consider the two statements

for any sets A and B, if $\mathcal{P}(A) = \mathcal{P}(B)$, then $A = B$;

and

there exists a set C such that $\mathcal{P}(C) = \emptyset$.

◯ *They are both true.*

✓ *Only the first is true.*

◯ *Only the second is true.*

◯ *They are both false.*

For any set C it is the case that $C = \bigcup_{c \in \mathcal{P}(C)} c$. It immediately follows that if $\mathcal{P}(A) = \mathcal{P}(B)$ for two sets A and B, then $A = B$. Thus the first statement is true. One can also say that if $A \neq B$, then $\mathcal{P}(A) \neq \mathcal{P}(B)$, and by contraposition that means that if $\mathcal{P}(A) = \mathcal{P}(B)$ then $A = B$.

For any set C it is the case that $\emptyset \subseteq C$ (because $\forall x\ x \in \emptyset \to x \in C$), so that $\emptyset \in \mathcal{P}(C)$ and thus $\mathcal{P}(C) \neq \emptyset$. Therefore there is no set C such that $\mathcal{P}(C) = \emptyset$ and the second statement is false.

It follows that the second circle must be ticked.

Exercise 8. Let $X = \{1, 2, 3, 4, 5\}$ and let $\mathcal{P}(X)$ denote the power set of X. Given the statements

$$\emptyset \in \mathcal{P}(X) \qquad\qquad \{\emptyset\} \in \mathcal{P}(X)$$

◯ Only the first is true.

◯ They are both true.

◯ Only the second is true.

◯ They are both false.

The power set $\mathcal{P}(X)$ is defined as the set that has all subsets of X as its elements. Furthermore, the statement

$$\forall x \in \emptyset\ (x \in X)$$

consists of a universal quantifier that ranges over an empty set and is thus true. According to the definition of "subset" it follows that $\emptyset \subseteq X$ so that it follows, using the definition of $\mathcal{P}(X)$, that $\emptyset \in \mathcal{P}(X)$.

If $\{\emptyset\} \in \mathcal{P}(X)$, then (using the definition of $\mathcal{P}(X)$), the set $\{\emptyset\}$ must be a subset of X, implying (according to the definition of a subset) that all elements of the set $\{\emptyset\}$ must also be elements of X. The set $\{\emptyset\}$ has just a single element, namely $\emptyset$, and $\emptyset$ is not one of the elements of X, because X just consists of the elements 1, 2, 3, 4, and 5. Thus $\{\emptyset\} \notin \mathcal{P}(X)$ and only the first answer is correct.

[2] We are not in the same situation as with f_δ, there is no contradiction here.